

For each question, show all related work and explain the solution. Giving only a correct solution with no or incomplete work, no explanation and/or improper notation can result in a score of 0 for that question.

1. 5 pts each - Suppose a “word” is any 7 letters of the alphabet with repeats allowed. Find the number of “words” with the following characteristics. Be sure to show your calculation and solution for each.
 - a. The total number of “words” (no restrictions)
 - b. How many “words” begin with the letter “A”
 - c. How many “words” begin with “S” and end with “F”
 - d. How many “words” begin with “ABC” in some order?
 - e. How many “words” have exactly one vowel (A, E, I, O, U)?
2. 10 pts - The given solution is incorrect. Find the error, explain why it is incorrect and how you would correctly solve the problem. The correct solution should relate to the incorrect solution. Show any calculations necessary.

Question: Find the number of ways to get two pairs of two different types of cards (like 2 3's and 2 7's) in a hand of 4 cards from a standard 52-card deck.

Incorrect Solution: There are 13 choices for the type of card for the first pair and $\binom{4}{2}$ ways to choose the cards in that pair. There are 12 choices left for the type of card in the second pair and $\binom{4}{2}$ ways to choose the cards in that pair. Therefore, the total number of ways to get two pair is:

$$13 \cdot \binom{4}{2} \cdot 12 \cdot \binom{4}{2}$$

3. 10 points - What is the probability that a positive integer less than 100 picked at random has all distinct digits?
4. 10 points - Suppose that two cards are drawn without replacement from a well-shuffled deck. What is the probability that both cards have numbers and that the numbers on the cards are the same (note that only the numbers 2 through 10 are shown on cards, since aces, kings, queens, and jacks are represented by letters).
5. 10 points - A pair of dice numbered 1-6 is loaded so that on the first die, the probability of a 3 appearing is $\frac{3}{8}$, with each other outcome appearing with a probability of $\frac{1}{8}$. On the second die, the probability of a 4 appearing is $\frac{2}{7}$, with each other outcome having a probability of $\frac{1}{7}$. What is the probability that the sum of the numbers rolled is equal to 7?
6. 10 points - Suppose $|A| = |B| = |C| = 100$, $|A \cap B| = 60$, $|A \cap C| = 50$, $|B \cap C| = 40$, and $|A \cup B \cup C| = 175$. Find $|A \cap B \cap C|$.

7. 15 points - Find a recurrence relation and initial condition for the number of fruit flies in a jar if there are 12 flies initially and every week there are six times as many flies in the jar as there were the previous week. Then find the explicit function.
8. 10 points - In a survey of 280 teenagers, it is found that 85 like carrots, 47 like peas, 74 like cauliflower, 22 like both carrots and peas, 32 like both peas and cauliflower, 42 like both carrots and cauliflower and only 12 like all 3 vegetables. How many teenagers don't like any of the vegetables?